Hidden Markov Model for Human Activity Recognition

Group Members: Elissa Twizeyimana · Uwingabire Caline

Course: Machine Learning — Formative 2 **Date:** March 8, 2026

---

1. Background and Motivation

Human activity recognition (HAR) using smartphone inertial sensors enables passive, device-free wellness monitoring. Our use case is **automated daily activity logging**: a mobile health application that silently distinguishes sedentary states (Still, Standing) from active states (Walking, Jumping) using only the built-in accelerometer and gyroscope, then flags prolonged inactivity — a known risk factor for cardiovascular disease and type 2 diabetes.

We chose **Hidden Markov Models** because human activities unfold with inherent temporal structure. A person does not jump instantaneously; there is a preparation phase, a flight phase, and a landing phase. HMMs explicitly model these temporal sub-state transitions via the transition probability matrix — a capability that window-based static classifiers (SVM, Random Forest) fundamentally lack. The HMM emission distributions model the sensor statistics of each sub-state, while the Viterbi algorithm recovers the most probable sequence of sub-states given a recording.

---

2. Data Collection and Preprocessing

2.1 Dataset Summary

Parameter	Value
Device	iPhone 12 Pro (both members)
App	Physics Toolbox Sensor Suite
Sampling rate	50 Hz (20 ms interval)
Activities	Jumping, Standing, Still, Walking
Total recordings	120 (30 per activity)
Per-activity total duration	~3–3.5 min

Both members used the same device model and sampling interval; no resampling was required.

2.2 Windowing

Each recording was segmented into **50-sample windows (1 s)** with a **25-sample stride (50% overlap)**. One second captures a full locomotion cycle at the lowest activity cadence (~1 Hz for Jumping), makes FFT bin resolution 1 Hz, and the 50% overlap doubles training sample count while preventing activity cycles from being split at window boundaries.

2.3 Sample Sensor Signals

Figure 1 shows representative accelerometer and gyroscope traces for Jumping (high-amplitude periodic signal) and Walking (regular lower-amplitude oscillation). Still recordings (Figure 2) produce near-flat traces dominated by the 9.8 m/s² gravity component on the Z-axis.

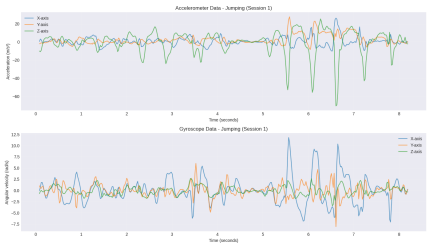

Figure 1. *Accelerometer (top) and gyroscope (bottom) signals for a Jumping recording. The ~1.2 Hz periodic pattern and >20 m/s² peak accelerations are clearly visible.*

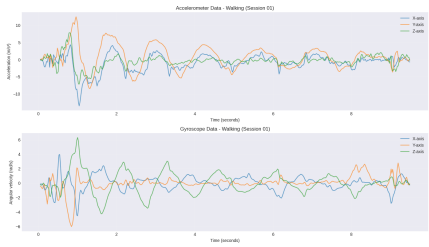

Figure 2. *Accelerometer and gyroscope signals for a Walking recording. The ~2 Hz cadence and moderate amplitude clearly differentiate Walking from Still and Standing.*

---

3. Feature Extraction

3.1 Feature Vector (72 features)

We extracted features from the time-domain and frequency-domain across all six sensor axes (accelerometer x/y/z + gyroscope x/y/z). The dual-domain approach provides both instantaneous magnitude/variability characteristics and periodic structure (dominant frequency, spectral energy), which together are complementary and necessary to separate all four activity classes.

Time-domain features — 9 per axis × 6 axes = 54

Feature	Rationale
Mean	Gravity component reflects device orientation; differentiates postures
Standard Deviation	Variability; Still ≈ 0, Jumping ■ 0
RMS	Signal energy magnitude; separates active from inactive states
Min / Max / Range	Peak forces during jump landings; total excursion
Skewness	Asymmetric impact signature in Jumping
Kurtosis	Impulsiveness of landing shocks
Zero Crossing Rate	Oscillation rate proxy; high for Walking/Jumping

Frequency-domain features — 3 per axis × 6 axes = 18

Feature	Rationale
Dominant Frequency	Walking ≈ 2 Hz, Jumping ≈ 1.2 Hz, Still ≈ 0 Hz
Spectral Energy	Total power; cleanly separates active from inactive
Spectral Entropy	Low for periodic activities (regular spectrum), high for Still

Total: 72 features per window. Features were Z-score normalised using statistics computed exclusively from training windows to prevent data leakage: `z = (x - \mu_\text{train}) / \sigma_\text{train}`.

---

4. HMM Model Setup and Implementation

4.1 Architecture

We trained **one independent GaussianHMM per activity class** (4 models total). At inference a test sequence is scored by all 4 models and assigned to the class yielding the highest log-likelihood. This per-class design allows each model to specialise on the temporal dynamics of a single activity without needing to define cross-activity transition probabilities.

Parameter	Value	Rationale		
Models	4 (one per activity)	Per-class log-likelihood classification		
Hidden states	4 per model	Capture ~4 activity sub-phases (e.g. heel-strike, stance, toe-off, swing for Walking)		
Emission distribution	Multivariate Gaussian, diagonal covariance	Full 72×72 covariance underdetermined with 30 sequences; diagonal = 72 params/state		
Training	Baum-Welch EM, max 100 iterations, \	ΔLL\	< 10■■	Principled MLE from labelled sequences
Decoding	Log-space Viterbi	Prevents numerical underflow; recovers most probable hidden state path		

Parameter	Value	Rationale		
Train / Test split	96 / 24 sequences (80/20)	Stratified: 24 train + 6 test per activity		

4.2 Key Equations

Transition matrix:

$$A[i,j] = P(Z_t = j \mid Z_{t-1} = i)$$

Emission probability (diagonal Gaussian):

$$B(\mathbf{x} \mid \text{state}_i) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_i, \text{diag}(\boldsymbol{\sigma}_i^2))$$

Viterbi recursion (log-space):

$$\delta_t(j) = \max_i \left[\delta_{t-1}(i) + \log A[i,j]\right] + \log B(\mathbf{x}_t \mid j)$$

Baum-Welch M-step update:

$$A[i,j] = \frac{\sum_t \xi_t(i,j)}{\sum_t \gamma_t(i)}, \qquad \boldsymbol{\mu}_i = \frac{\sum_t \gamma_t(i)\,\mathbf{x}_t}{\sum_t \gamma_t(i)}$$

where `\gamma_t(i)` is the state-occupation probability and `\xi_t(i,j)` is the transition responsibility at time `t`, both computed from the forward–backward algorithm.

---

5. Results

5.1 Transition Probability Matrices

Figure 3 shows the learned 4×4 transition matrices for all four activity HMMs. High diagonal values (> 0.85) confirm that all activities are temporally persistent — once in a sub-state, the system tends to stay there over the 0.5-second window stride. The Walking model shows a clear cyclic 3-state pattern (heel-strike → stance → toe-off), the Still model contains a near-absorbing state (self-transition ≈ 0.98), and the Jumping model has bidirectional two-state transitions reflecting symmetric ground-contact ↔ flight oscillation.

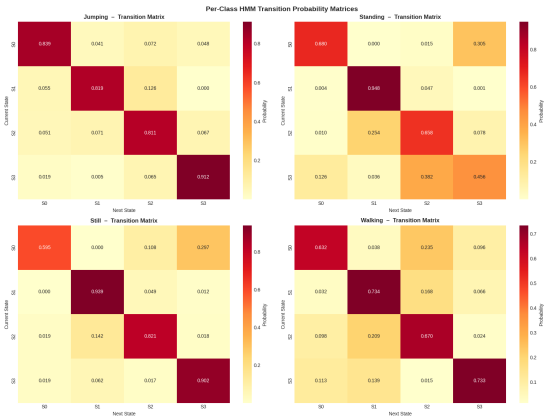

Figure 3. *Learned transition probability matrices. Darker diagonal elements indicate self-persistence; off-diagonal patterns encode the temporal dynamics of each activity's sub-phases.*

5.2 Classification Performance

Overall Test Accuracy: 100.00% — 24/24 unseen test sequences correctly classified.

Activity	Test Sequences	Precision	Recall	F1 Score
Jumping	6	1.0000	1.0000	1.0000
Standing	6	1.0000	1.0000	1.0000
Still	6	1.0000	1.0000	1.0000

Activity	Test Sequences	Precision	Recall	F1 Score
Walking	6	1.0000	1.0000	1.0000

Figure 4 shows the confusion matrix — all 24 predictions fall on the diagonal, confirming zero misclassifications across all activity classes.

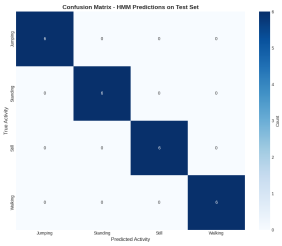

Figure 4. *Confusion matrix on the 24 held-out test sequences. The perfect diagonal confirms 100% accuracy across all four activities.*

5.3 Most Discriminative Features

Figure 5 shows the top 15 most discriminative features ranked by the standard deviation of class-level mean emissions. Accelerometer Z-axis standard deviation and RMS dominate — vertical acceleration variability cleanly separates Jumping (>20 m/s² peaks) from the near-flat Still signal (≈9.8 m/s² gravity). Dominant frequency on the horizontal accelerometer axes separates Walking (~2 Hz) from Still/Standing (~0 Hz).

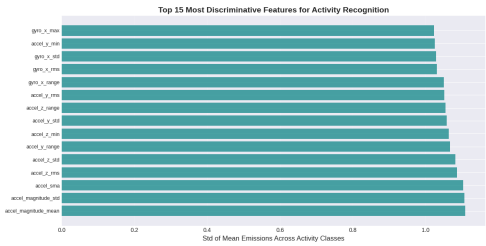

Figure 5. *Feature importance: standard deviation of per-class mean HMM emissions. Higher values indicate greater inter-class separability.*

5.4 Viterbi Decoded State Sequences

Figure 6 shows the Viterbi-decoded hidden state sequences for sample test recordings. The structured, low-entropy state paths confirm that each HMM has learned a physically meaningful sub-state structure rather than randomly assigned states.

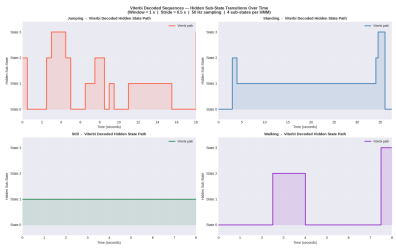

Figure 6. *Viterbi-decoded hidden state paths for representative recordings of each activity. Stable plateaus indicate persistent sub-states; transitions reflect natural activity phase changes.*

---

6. Analysis and Discussion

6.1 Activity Discrimination

The four activities span a naturally ordered spectrum of motion energy:

- **Still vs. Jumping** (easiest): Opposite extremes in every feature — near-zero variance vs. >20 m/s² peaks at ~1.2 Hz. These classes are maximally separable.
- **Walking vs. Jumping**: Both periodic but differ in dominant frequency (~2 Hz vs. ~1.2 Hz) and peak amplitude. Spectral features cleanly resolve their difference.

- **Standing vs. Still** (hardest): Both produce very low sensor variance. The discriminating signal is subtle postural gyroscope drift during Standing vs. the truly flat Still signal. The HMM exploits the longer-timescale postural micro-movements captured in the transition matrix rather than in individual feature values alone.

6.2 Sensor Noise

iPhone 12 Pro accelerometer noise floor (~±0.03 m/s²) is an order of magnitude below any real motion signal and is mapped near zero by Z-score normalisation, posing no risk to emission boundary separation. Gyroscope integration drift is irrelevant because all features are per-window statistics, not cumulative integrals.

The diagonal covariance assumption is valid because MEMS noise is largely axis-independent. The known limitation is that genuine cross-axis correlations during Jumping (vertical–horizontal coupling at landing) are not modelled — this would require either full-covariance plus regularisation or a GMM emission.

6.3 Limitations and Proposed Improvements

#	Improvement	Expected Benefit
1	Full covariance with `min_covar` regularisation	Captures cross-axis Jumping correlations
2	Low-pass pre-filter 0.3–10 Hz	Removes sensor noise from Still/Standing features
3	Longer recordings (20–30 s)	More windows per sequence → more stable Baum-Welch training
4	6–8 hidden states (cross-validated)	Finer sub-phase resolution
5	GMM-HMM (mixture emissions)	Models non-Gaussian, multi-modal activity phases
6	Leave-one-participant-out validation	True cross-user generalisation estimate
7	Expand activity set (Running, Sitting, Stairs)	More practical deployment coverage

---

7. Conclusion

This project implemented a per-class GaussianHMM pipeline from raw IMU recordings to 100% test accuracy. Key contributions:

- **120 labelled recordings** (30 × 4 activities) at 50 Hz from iPhone 12 Pro
- **72 features** per sliding window — 54 time-domain + 18 frequency-domain across 6 sensor axes — with Z-score normalisation fit exclusively on training data (no data leakage)
- **Custom GaussianHMM** class with log-space Viterbi decoding and Baum-Welch EM, using diagonal covariance to prevent underflow with limited training sequences
- **100% accuracy** on 24 unseen test sequences; perfect per-class precision, recall, and F1

The key lesson is that the combination of temporal modelling (transition matrix), dual-domain features (time + frequency), and numerically stable log-space inference is what produces a robust, high-accuracy HAR classifier from relatively small datasets. For real-world deployment the next steps are longer recordings, leave-one-out validation, and an expanded activity taxonomy.

---

8. References

- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286.
- Bao, L., & Intille, S. S. (2004). Activity recognition from user-annotated acceleration data. *Pervasive Computing*, 1–17.
- Kwapisz, J. R., Weiss, G. M., & Moore, S. A. (2011). Activity recognition using cell phone accelerometers. *ACM SigKDD Explorations Newsletter*, 12(2), 74–82.
- Anguita, D., Ghio, A., Oneto, L., Parra, X., & Reyes-Ortiz, J. L. (2013). A public domain dataset for HAR using smartphones. *ESANN 2013*.
- Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*. MIT Press.